

Problem Set 1

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Due: February 11, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Wednesday February 11, 2026. No late assignments will be accepted.

Question 1

The Kolmogorov-Smirnov test uses cumulative distribution statistics test the similarity of the empirical distribution of some observed data and a specified PDF, and serves as a goodness of fit test. The test statistic is created by:

$$D = \max_{i=1:n} \left\{ \frac{i}{n} - F_{(i)}, F_{(i)} - \frac{i-1}{n} \right\}$$

where F is the theoretical cumulative distribution of the distribution being tested and $F_{(i)}$ is the i th ordered value. Intuitively, the statistic takes the largest absolute difference between the two distribution functions across all x values. Large values indicate dissimilarity and the rejection of the hypothesis that the empirical distribution matches the queried theoretical distribution. The p-value is calculated from the Kolmogorov- Smirnov CDF:

$$p(D \leq d) = \frac{\sqrt{2\pi}}{d} \sum_{k=1}^{\infty} e^{-(2k-1)^2 \pi^2 / (8d^2)}$$

which generally requires approximation methods (see Marsaglia, Tsang, and Wang 2003). This so-called non-parametric test (this label comes from the fact that the distribution of the test statistic does not depend on the distribution of the data being tested) performs

poorly in small samples, but works well in a simulation environment. Write an R function that implements this test where the reference distribution is normal. Using R generate 1,000 Cauchy random variables (`rcauchy(1000, location = 0, scale = 1)`) and perform the test (remember, use the same seed, something like `set.seed(123)`, whenever you're generating your own data).

Before conducting the test, it is crucial to outline the Null and Alternative hypothesis.

Null Hypothesis: The datasets are the same or follow the same, specific distribution. Specifically, this dataset follows a Normal Distribution.

Alternative Hypothesis: The datasets differ and follow a different distribution. Specifically, this dataset does NOT follow a Normal Distribution.

```
1 set.seed(123)
2 data <- (rcauchy(1000, location = 0, scale = 1))
3 #creating the function
4 KSpval <- function(data) {
5   k <- length(data) #initializing for iteration.
6   ECDF <- ecdf(data)
7   empiricalCDF <- ECDF(data)
8   d_value <- max(abs(empiricalCDF - pnorm(data))) #test statistic
9   i <- 1:k #taking a vectorized approach instead of a loop for calculating the
      terms
10  terms<- exp(-(((2*i - 1)^2) * pi^2)/(8*(d_value^2))) #calculating the terms
      for all 100 k's
11  d_obs <- sqrt(2 * pi)/d_value * sum(terms) #calculating the observation
12 return(d_obs)
13 }
14 KSpval(data)
15 # [1] 5.652523e-29
```

After conducting the test, we have obtained a p-value of 5.652523e-29, which, at a threshold of $\alpha = 0.05$, is extremely statistically significant. This means we have sufficient evidence to reject the null hypothesis that our dataset follows a normal distribution.

Question 2

Estimate an OLS regression in R that uses the Newton-Raphson algorithm (specifically BFGS, which is a quasi-Newton method), and show that you get the equivalent results to using `lm`. Use the code below to create your data.

```
1 linear.lik <- function(theta, y, X) { #creating a log of likelihood function
2   n <- nrow(X)
3   k <- ncol(X)
4   beta <- theta[1:k]
5   sigma2 <- theta[k+1]^2
6   e <- y - X%*%beta
7   logl <- -.5*n*log(2*pi) - .5*n*log(sigma2) - ( (t(e) %*%e) / (2*sigma2) )
8   return(-logl)
9 }
10 #running with optim.
11 MLEt <- optim(fn=linear.lik, y=data$y, X = cbind(1, data$x), par=c(1,1,1),
12             hessian=T, method = "BFGS")
13 MLEt$par
14 #[1] 0.1429829 2.7263116 -1.4423360
15 linear <- summary(lm(data$y ~ data$x))
16 # (Intercept) 0.13919
17 # data$x      2.72670
```

To ensure the results from the linear likelihood function are correct and that we are correctly estimating an OLS regression, we can compare them to R's `lm` function. After running both, we see that the intercept values differ slightly, but when rounded to the second decimal point, are equivalent. The beta values are 2.72. The intercept values are 0.14. There will be a slight difference, since the OLS estimator of variance is unbiased, whereas the ML estimator is biased. If we were to have a larger sample, however, the differences would be insignificant.